

Piyush Mohan Bhattarai

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Data analyst and ML practitioner with experience building scalable analytics systems, predictive models, and BI dashboards. Proven track record of translating complex business problems into data-driven solutions, including reducing churn, cutting delivery timelines, and improving operational efficiency. Experienced in cross-functional stakeholder management, Agile delivery, and ML workflows from EDA through model evaluation and operational adoption.

SKILLS

Analytics & Modeling: Python, Pandas, NumPy, Scikit-learn, R, SQL, EDA, ETL Pipelines, Predictive Analysis, Time-Series Forecasting, LSTM, ARIMA, Statistical Modeling, Data Warehousing

Machine Learning & NLP: Random Forest, Tree-Based Models, Deep Learning, Text Embeddings, LLM-Based NLP, RAG Pipelines, A/B Testing, Model Evaluation & Deployment

Business Intelligence: Tableau, Power BI, Looker Studio, Excel, Power Query, DAX, Pivot Tables, VLOOKUP, XLOOKUP, SAP, Google Sheets

Business & Delivery: Stakeholder Management, Agile Delivery, KPI Monitoring, Process Optimization, ERP/CRM Systems, Cross-Functional Coordination, UAT, Data Governance

WORK EXPERIENCE

SecurityPal Inc.

June 2022 – July 2024

Lead Data Analyst

- Built and scaled NLP-based analytics workflows using Python, SQL, and text embedding models to extract insights from unstructured security questionnaire data, improving customer operations efficiency by 20%.
- Automated data pipelines integrating customer health, product usage, and feedback data, enabling adoption trend analysis, engagement pattern monitoring, and early risk detection to support customer success decisions.
- Designed and executed A/B tests on ML model variants using predefined KPIs and structured feedback from 50+ analysts, driving data-driven model selection and continuous workflow improvement.
- Implemented LSTM and ARIMA time-series forecasting models to anticipate questionnaire volume and turnaround time, improving resource allocation, capacity planning, and analyst bandwidth management.
- Consolidated 50,000+ questions from the Knowledge Library to develop SP-Core, a standardized security questionnaire framework for third-party vendor risk assessment, reducing average turnaround time by 30%.
- Built Tableau dashboards and client journey visualizations, with timeline prototypes designed in Figma, to identify operational trends, service gaps, and workflow inefficiencies.
- Led a team of 10 analysts, coordinating cross-functional reporting workflows and reducing BAU delivery timelines by 25% through structured process optimization and KPI tracking.

Research and Innovation Unit

December 2020 – June 2022

Research Analyst

- Developed a customer churn prediction model using Random Forest and SQL-based feature engineering, identifying at-risk segments and reducing churn by 15% over 3 months.
- Conducted sentiment analysis and exploratory data analysis on customer feedback datasets to uncover key churn drivers, enabling more targeted retention strategies and improved customer segment identification.
- Built Tableau dashboards visualizing churn trends and customer risk segments, enabling stakeholders to monitor retention KPIs and act on early warning indicators.
- Authored research on Digital Twin applications in 3D printing and Building Information Modeling (BIM), covering planning, coordination, and resource optimization use cases.

- Led requirements gathering and stakeholder alignment across multiple software projects, translating business needs into functional specifications and workflow maps that guided development and reduced scope creep.
- Analyzed operational and service delivery data using SQL and BI reporting tools to identify process inefficiencies, contributing to improvements that reduced delivery bottlenecks by 15%.
- Defined data capture requirements and collaborated with engineering teams to ensure accurate data structure and downstream usability for reporting and analytics.
- Designed UAT test cases, tracked defects across business and technical teams, and validated system outputs against requirements to improve delivery quality.

- Led stakeholder analysis and requirements gathering for Nepal Telecom’s 4G LTE Goods Tracking System (GTS), reducing requirement-related project delays by 25% through structured documentation and cross-team alignment.
- Evaluated and mapped operational workflows to functional specifications in an Agile environment, ensuring system capabilities aligned with business objectives.
- Managed UAT and defect tracking across business and technical teams, reducing testing cycle time by 30% during Provisional Acceptance Testing (PAT).
- Maintained GTS database integrity and reporting consistency, providing reliable operational visibility for project tracking and executive decision-making.

EDUCATION

GPA: 4.00

Relevant Projects:

- Decision Support for Program Selection (Human–AI Interaction using Phi-3 Mini) – Built an interactive LLM-based tool for admissions/advising teams to explore program fit by simulating diverse student personas and generating rationale-based responses to program options.
- Built and compared CNN and ResNet-18 architectures for multi-class chest X-ray classification, evaluated model performance, and deployed the final model through a Gradio interface.
- Fine-tuned DistilBERT on the Banking77 dataset for multiclass intent classification of customer support queries, and built a Gradio-based prototype for support-triage demonstration.

CERTIFICATIONS

Google AI Professional Certificate | Machine Learning Specialization (Stanford Online) | Google Data Analytics Professional Certificate | Generative AI Fundamentals (Databricks) | Six Sigma White Belt (Educate 360)